

Reading Notes: "Design Meets Neuroscience: A Preliminary Review of Design Research Using Neuroscience Tools"

Hu, L., & Shepley, M. M. (2022). Design meets neuroscience: a preliminary review of design research using neuroscience tools. *Journal of interior design*, 47(1), 31-50.

I. Background & Objective

- **Context:** Traditional design research methods (interviews, observations) struggle to capture non-verbal and subconscious cognitive processes. Recent advances in neuroscience tools (EEG, fMRI, NIRS, PET) offer a direct way to investigate how brain function supports design thinking.
- **Goal:** To provide a preliminary review of how design researchers have used neuroscience tools, focusing on: (1) explored neural research topics, (2) available tools, their contributions and limitations, and (3) the challenges of this research.

II. Core Neuroscience Tools Overview

- **Key Trade-off: Temporal resolution** (when) vs. **Spatial resolution** (where).
 - **fMRI:** High spatial resolution (mm), answers "**where**" questions. Limitations: low temporal resolution, high cost, noisy, immobile subjects.
 - **EEG:** High temporal resolution (ms), answers "**when**" questions. Limitations: poor spatial resolution, but is portable and more affordable.
 - **NIRS (fNIRS):** Portable, tolerant of motion, a good alternative to fMRI. Limitations: poorer spatial and temporal resolution.
 - **PET:** High sensitivity, quiet environment. Limitations: involves radioactive substances, safety concerns limit use.
- **Recommendation:** Tool choice depends on the research question. Acknowledging technical limitations is critical, and future research should consider multimodal imaging.

III. Neural Correlates of Design Thinking & Creativity

- **Design Thinking vs. Problem-Solving:** Alexiou et al. (2009) fMRI study showed that ill-defined design tasks (e.g., "make the room comfortable") activate a more extensive neural network than well-defined problem-solving tasks, including the **dorsolateral prefrontal cortex (DLPFC)**, **anterior cingulate cortex (ACC)**, and **medial temporal lobe**. This provides neural evidence for design as a distinct cognitive mode.
- **Creativity Studies:**
 - **Visual Creativity:** The **prefrontal cortex** and **occipitotemporal regions** are crucial for generating and manipulating visual imagery.
 - **Generation vs. Evaluation:** Ellamil et al. (2012) fMRI study found that **creative generation** primarily activates the **medial temporal lobe memory network**, while **creative evaluation** involves the joint activation of the **executive network** and the **default mode network**.

- **Training & Originality:** Kowatari et al. (2009) showed that originality in design students correlates with **right prefrontal cortex** dominance, and professional training alters brain activation patterns.
- **Key Brain Regions:** Across studies, the **prefrontal cortex (especially DLPFC)**, **anterior cingulate cortex (ACC)**, and **parietal cortex** are consistently involved in design and creative cognition.

IV. Neural Correlates of Design Evaluation & Aesthetics

- **Applications:** User interface design, packaging, product design, architecture, and interior design.
- **Consumer Neuroscience:** Neural responses (e.g., from EEG/fMRI) can predict consumer preferences for marketing stimuli like brands and packaging, revealing subconscious preferences.
- **Interior/Architecture Applications:**
 - **Visual Elements:** **Curvilinear contours** activate the **ACC** (related to aesthetic judgment) more than rectilinear ones. Lighting type (direct/indirect) affects emotion and EEG signals.
 - **Non-Visual Elements:** Sound, ceiling height, and spatial orientation influence wayfinding and spatial experience.
 - **Neuroaesthetics:** The study of aesthetic perception at the neural level. EEG studies show that **unconscious color preferences** influence attention-related brain activity. fMRI studies show that **most and least favorite colors** elicit different brain activation patterns.

V. Major Challenges

- **Technical Limitations:** The inherent trade-off between temporal and spatial resolution for each tool.
- **Practical Difficulties:** High equipment cost, operational complexity, limited accessibility, and strict participant recruitment criteria (e.g., no metal implants for fMRI).
- **Experimental Challenges:** Isolating a single cognitive process in a lab is difficult. Ensuring ecological validity is a concern. Individual differences (e.g., medical states, drugs) are confounding variables.

VI. Conclusion & Future Directions

- **Value:** Neuroscience tools provide objective data that complements self-reports and behavioral measures, offering new insights into the human experience of design.
- **Recommendation:** Design researchers must understand the trade-offs and challenges of each tool. **Collaboration with neuroscientists is highly recommended** to design valid protocols. Future work should focus on developing more ecologically valid experimental paradigms (e.g., fMRI-compatible drawing tablets) and multimodal imaging.